

A combinatorial proof of the M-convexity of cylindric skew Schur polynomials

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Title: A combinatorial proof of the M-convexity of cylindric skew Schur polynomials.

Provenance, stated by kind rather than as a blanket “nothing has changed”, because a blanket claim is a universally quantified assertion about every other line and any later edit anywhere refutes it. Relative to

68184eb, the mathematics HAS changed, three times, and each change is recorded in the text at the point where it applies: (M1) 23 September 2026, cycle 2 — a false clause in the Gaps section replaced by Proposition 5.4, which is proved; see (iv) below. (M2) 25 September 2026, cycle 1 — an erratum to the proof of Proposition 6.3: one displayed inequality ran the wrong way, and the count is now made on the complement of S^ . The statement is unchanged. Remark 6.4 records it in full. (M3) 25 September 2026, cycle 2 — additions, no retraction: Lemma 6.2 (previously a one-sentence justification inside the proof of Proposition 6.3), the new §8 (Theorems 8.1 and 8.4, closing Gaps item 2), a half-line of periodicity in the proof of Corollary 3.4, and a tightened hypothesis in Lemma 5.2 (parts (1) and (3) are unconditional). Renumbering caused by (M3), which invalidates numeric citations to the earlier build: Proposition “6.2” (the polymatroid exchange) is now Proposition 6.3, and the erratum remark “6.3” is now Remark 6.4. Theorem 7.1, Lemma 4.1, Proposition 4.2 and Corollary 3.4 keep their numbers. Cite by label where possible.*

Non-mathematical corrections, none of which alters a statement, proof, definition or numerical claim:

(i) the cover block itself; (ii) 23 September 2026, cycle 1 — four missing references added to the bibliography (Gessel–Krattenthaler, Postnikov, McNamara, Neyman), with two sentences in §1 and one clause in §8 amended to cite them; (iii) 23 September 2026, cycle 2 — bibliography provenance repaired. The entry [3] carried a title that is not the title of any paper; the correct title, journal coordinates and DOI are now given, verified against two independent bibliographic sources. The annotations “Not read at source” on [5], [6] and [7] were true when written and became false on 23 September 2026, when all three were obtained and read; each now records what was read, from which copy, and with what locators. (v) 26 September 2026 — four sentences asserting that Lemma 6.2 is not formalised in Lean were true when written on 25 September, cycle 2, and were falsified the same evening by the Lean development that closed exactly that gap. They are corrected here and in Remark 6.4. The surviving Lean caveat — Murota’s symmetric exchange axiom — is unchanged and is stated in the same remark. A cover line naming the commit that this build corresponds to has also been added, as PROTOCOL 2.3 requires; the block previously named only the first commit.

Detail of (M1): *(iv) 23 September 2026, cycle 2 — Gaps item (iii) of the Gaps section (which is §11, not §8 as an earlier cover block said) asserted that $\hat{\lambda}(\lambda/\mu, \ell)$ “grows with ℓ until ℓ reaches the minimal chain length”. That assertion is false; the recursion defining the greedy chain contains no ℓ . The clause has been replaced by Proposition 5.4 ($\hat{\lambda}$ is independent of ℓ on its whole domain), which is proved. No theorem, lemma, definition or numerical claim of §§1–7 is affected by (M1): $\hat{\lambda}$ is used only for a fixed ℓ throughout, and the false clause appeared only in the list of gaps.*

Abstract

Wang, Zhang and Zhang [WZZ, arXiv:2401.14632, Problem 5.2] ask for a combinatorial proof of the M-convexity of cylindric skew Schur polynomials *based on their tableau interpretation*. Their own Corollary 4.9 obtains M-convexity, but only by importing two external results — Lam’s Corollary 8.5 (affine Stanley symmetric functions expand positively in dual k -Schur functions) and Lee’s Corollary 5 — and the sibling Problem 5.1 makes it explicit that what is wanted is a proof that does *not* route through Lam’s theorem.

We give such a proof. Working in the periodic Maya-diagram (bead) model, a semistandard cylindric skew tableau becomes a chain of n -periodic bead configurations, and its weight becomes the increment sequence of the single scalar statistic $u^t = \sum_{i=1}^m x_i^t$. The entire argument then rests on one observation (Lemma 3.1): *with its two neighbours held fixed, a row of the chain varies independently in each bead, over a product of integer intervals $\prod_i [L_i, U_i]$, whose endpoints satisfy the identity $\sum_i (L_i + U_i) = \sum_i (p_i + r_i)$* (Lemma 3.2). From this we obtain, in turn, a cylindric Bender–Knuth involution, the exchange move (a single bead sliding one site), the dominance-ideal property of the weight set, and — via a greedy chain — the unique dominance maximum. A self-contained polymatroid argument converts this into M-convexity.

The proof cites no external theorem: not Lam’s Corollary 8.5, not Lee’s Corollary 5, not Postnikov’s symmetry theorem for cylindric skew Schur functions (which we reprove along the way, Corollary 3.4), and not Rado’s theorem — though in this last case the honest statement is that we *reprove* the direction we need, from Lemma 4.1, rather than avoid it (Remark 8.5). It also produces more than M-convexity: it identifies the Newton polytope explicitly as the $\hat{\lambda}$ -permutahedron for a partition $\hat{\lambda}$ computed by an explicit greedy algorithm.

1 Introduction

1.1 The problem

Let $f = \sum_{\alpha} c_{\alpha} x^{\alpha} \in \mathbb{R}[x_1, \dots, x_{\ell}]$ and let $\text{supp}(f) = \{\alpha : c_{\alpha} \neq 0\}$. A set $J \subseteq \mathbb{N}^{\ell}$ is *M-convex* if for all $\alpha, \beta \in J$ and every index i with $\alpha_i > \beta_i$ there is an index j with $\alpha_j < \beta_j$ and $\alpha - e_i + e_j \in J$. A polynomial is M-convex if its support is.

Cylindric partitions are due to Gessel and Krattenthaler [6]; cylindric skew Schur functions were introduced by Postnikov [5], who proved that they are symmetric, and were studied further by McNamara [7]. Wang–Zhang–Zhang (henceforth WZZ) prove in [1, Cor. 4.9] that every cylindric skew Schur polynomial is M-convex. Their route is: M-convexity of dual k -Schur polynomials (their Theorem 3.6, a genuinely combinatorial theorem about k -Kostka numbers) $\Rightarrow$ M-convexity of affine Stanley symmetric polynomials (their Theorem 4.4, which invokes [2, Cor. 8.5]) $\Rightarrow$ M-convexity of cylindric skew Schur polynomials (their Cor. 4.9, which invokes [3, Cor. 5]).

They then pose, verbatim:

Problem 1.1 (WZZ, Problem 5.2). Find a combinatorial proof of the M-convexity of cylindric skew Schur polynomials based on their tableau interpretation.

The motive is stated one paragraph earlier, attached to the sibling Problem 5.1: *“One of the key ingredients of our approach is Theorem [Lam, Cor. 8.5], a deep result obtained by Lam. It would be interesting to find a direct proof of Theorem [Thm. 4.4] based on the definition given by (4.1).”* Problem 5.2 is posed as “similar to Problem 5.1” and inherits this motive. So the object of the exercise is a proof that bypasses the import chain. Since the truth of the statement is not in question, the success criterion is checkable: *if the argument still consumes Lam’s Corollary*

8.5 or Lee’s Corollary 5, it has not answered the problem. We therefore state explicitly, in §9, every external result the argument uses. The list is empty.

1.2 What we prove

Theorem 1.2 (Main theorem; see Theorem 7.1). *Let λ/μ be a cylindric skew shape in $\mathcal{C}^{n,m}$ and let $\ell \geq 1$. If the cylindric skew Schur polynomial $s_{\lambda/\mu}^c(x_1, \dots, x_\ell)$ is nonzero, then there is a partition $\hat{\lambda} = \hat{\lambda}(\lambda/\mu, \ell)$ of $|\lambda/\mu|$, computed by an explicit greedy algorithm, such that*

$$\text{supp}(s_{\lambda/\mu}^c(x_1, \dots, x_\ell)) = \{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \trianglelefteq \hat{\lambda}\} = \mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^\ell.$$

In particular $s_{\lambda/\mu}^c$ is M-convex, is SNP, and $\text{Newton}(s_{\lambda/\mu}^c) = \mathcal{P}_{\hat{\lambda}}$ is a $\hat{\lambda}$ -permutahedron.

This is strictly stronger than WZZ’s Corollary 4.9: it names the polytope.

1.3 The shape of the argument, and why it is short

The tableau interpretation of $s_{\lambda/\mu}^c$ is a chain $\mu = \lambda^0 \subseteq \lambda^1 \subseteq \dots \subseteq \lambda^\ell = \lambda$ of cylindric shapes with cylindric-horizontal-strip steps, weighted by $\alpha_t = |\lambda^t/\lambda^{t-1}|$. In the periodic Maya-diagram model a cylindric shape becomes a strictly increasing $x : \mathbb{Z} \rightarrow \mathbb{Z}$ with $x_{i+m} = x_i + n$ (§2), and two things happen at once:

- (i) **The weight linearises.** Putting $u^t = \sum_{i=1}^m x_i^t$, one has $\alpha_t = u^t - u^{t-1}$. The weight of a cylindric tableau is the increment sequence of a single integer statistic. Consequently $\alpha_1 + \dots + \alpha_r = u^r - u^0$ telescopes, which is what makes the greedy argument of §5 work.
- (ii) **The chain decouples.** The condition that $\lambda^{t-1} \subseteq \lambda^t$ and $\lambda^t \subseteq \lambda^{t+1}$ are both horizontal strips involves x_i^t only through x_i^{t-1} and x_i^{t+1} — never through x_j^t for $j \neq i$. So with its neighbours fixed, the row x^t ranges over a box $\prod_i [L_i, U_i]$ (Lemma 3.1).

Everything else is bookkeeping on that box. The one identity that has to be checked is

$$\sum_{i=1}^m (L_i + U_i) = \sum_{i=1}^m (p_i + r_i), \quad p = x^{t-1}, \quad r = x^{t+1},$$

and its proof (Lemma 3.2) is “max + min = sum” plus one wrap-around term. It is worth flagging that the wrap-around contributes exactly $+n$, and that without it the identity would read $\sum (L_i + U_i) = \sum (p_i + r_i) - n$. *The cylindric case works because of the periodicity, not in spite of it.* (I record this because I first derived the wrong version, by dropping the wrap-around term; the computation in §10 caught it.)

1.4 Relation to the ribbon/Maya move of Naprienko

Naprienko [4, Lemma 4.4] records that if λ/μ is a *ribbon* with minimal content k and maximal content K , then $\alpha_0(\lambda) = (\alpha_0(\mu) \setminus \{k\}) \cup \{K + 1\}$ — one bead hops from site k to site $K + 1$, possibly over other beads. That is a natural candidate for an “exchange move”. It is not the move used here, and the distinction matters:

- a **ribbon** of size r is one bead hopping r sites, jumping freely over occupied sites;

- a **horizontal strip** of size r is several beads hopping right with pairwise disjoint closed trajectories — no bead may pass or land on another (Lemma 2.6).

Cylindric skew Schur functions are built from horizontal strips, so the ribbon move is the wrong local move for this problem. The move that actually implements the M-convex exchange axiom is smaller still: *one bead in one row of the chain slides one site to the left* (Corollary 3.5).

2 The bead model

Fix integers $n > m \geq 1$.

Definition 2.1. A *cylindric shape of type (n, m)* is a strictly increasing map $x : \mathbb{Z} \rightarrow \mathbb{Z}$, written $i \mapsto x_i$, satisfying $x_{i+m} = x_i + n$ for all $i \in \mathbb{Z}$. We write $\mathcal{C}^{n,m}$ for the set of these. A shape is determined by the tuple $(x_1, \dots, x_m)$, which satisfies $x_1 < x_2 < \dots < x_m < x_1 + n$. Set

$$u(x) = \sum_{i=1}^m x_i.$$

For $\nu, \rho \in \mathcal{C}^{n,m}$ write $\nu \subseteq \rho$ if $\nu_i \leq \rho_i$ for all i , and in that case put $|\rho/\nu| = u(\rho) - u(\nu)$. Write $\nu \ll \rho$ (“ ρ/ν is a horizontal strip”) if

$$\nu_i \leq \rho_i < \nu_{i+1} \quad \text{for all } i \in \mathbb{Z}. \quad (1)$$

By periodicity it suffices to check (1) for $i = 1, \dots, m$, reading $\nu_{m+1} = \nu_1 + n$.

2.1 Dictionary to the lattice-path definition

We recall the definition used by WZZ [1, §5], following Lam and Postnikov. A cylindric shape is an infinite lattice path in $\mathbb{Z}^2$ with steps upwards and to the left, invariant under the shift $(m-n, m)$, where $1 \leq m \leq n-1$; $\mu \subseteq \lambda$ means μ lies weakly to the left of λ ; and a *semistandard cylindric skew tableau* of shape λ/μ and weight $\alpha = (\alpha_1, \dots, \alpha_\ell)$ is a chain $\mu = \lambda^0 \subseteq \lambda^1 \subseteq \dots \subseteq \lambda^\ell = \lambda$ of cylindric shapes such that each λ^t/λ^{t-1} contains at most one box in each column and α_t boxes in any $n-m$ consecutive columns. Then $s_{\lambda/\mu}^c = \sum_T x^{\text{weight}(T)}$.

Proposition 2.2. Index the rows of the cylinder by $i \in \mathbb{Z}$, increasing downwards, and for a cylindric shape P let $R_i(P) \in \mathbb{Z}$ be the column index of the right-hand boundary of row i (equivalently, the position of the vertical boundary segment of P at height i). Then R_i is weakly increasing in i , and the shift $(m-n, m)$ gives $R_{i+m} = R_i + (n-m)$. Set

$$x_i(P) := R_i(P) + i. \quad (2)$$

Then $P \mapsto (x_i(P))_{i \in \mathbb{Z}}$ is a bijection from cylindric shapes onto $\mathcal{C}^{n,m}$, and under it:

1. $\mu \subseteq \lambda$ (weakly to the left) $\iff \mu_i \leq \lambda_i$ for all i ;
2. the number of boxes of λ/μ in a fundamental domain for the shift — m consecutive rows, equivalently $n-m$ consecutive columns — equals $u(\lambda) - u(\mu)$;
3. λ/μ has at most one box in each column $\iff \mu \ll \lambda$.

Proof. R_i is weakly increasing because the path has only up and left steps, so the right-hand boundary of row $i + 1$ is weakly to the right of that of row i ; and the shift moves m rows down and $n - m$ columns right, giving $R_{i+m} = R_i + (n - m)$. Hence $x_{i+m} - x_i = (n - m) + m = n$ and $x_{i+1} - x_i = (R_{i+1} - R_i) + 1 \geq 1$, so $(x_i) \in \mathcal{C}^{n,m}$. Conversely $R_i = x_i - i$ recovers P , and R so obtained is weakly increasing with $R_{i+m} = R_i + (n - m)$, so every element of $\mathcal{C}^{n,m}$ arises. This is the bijection.

(i) μ lies weakly to the left of λ iff $R_i(\mu) \leq R_i(\lambda)$ for every row i , which by (2) is $\mu_i \leq \lambda_i$.

(ii) Row i of λ/μ occupies the columns $R_i(\mu) + 1, \dots, R_i(\lambda)$, so it contains $R_i(\lambda) - R_i(\mu) = \lambda_i - \mu_i$ boxes. Rows $1, \dots, m$ form a fundamental domain for the shift, which sends row i to row $i + m$; so does any block of $n - m$ consecutive columns, which the shift moves by $n - m$. Both therefore contain $\sum_{i=1}^m (\lambda_i - \mu_i) = u(\lambda) - u(\mu)$ boxes.

(iii) By (ii), rows i and $i + 1$ occupy the column intervals $[R_i(\mu) + 1, R_i(\lambda)]$ and $[R_{i+1}(\mu) + 1, R_{i+1}(\lambda)]$, and since R is weakly increasing these intervals are in nondecreasing position; they are disjoint iff $R_{i+1}(\mu) + 1 > R_i(\lambda)$, i.e. $R_{i+1}(\mu) \geq R_i(\lambda)$, i.e. $x_{i+1}(\mu) - (i + 1) \geq x_i(\lambda) - i$, i.e. $x_i(\lambda) < x_{i+1}(\mu)$. If rows i and $i + 1$ are disjoint then so are rows i and $i + k$ for every $k \geq 2$, the intervals being ordered. Hence no column carries two boxes iff $\lambda_i < \mu_{i+1}$ for all i ; together with $\mu_i \leq \lambda_i$ from (i) this is exactly (1). $\square$

Remark 2.3 (Fixing the orientation). The sign in (2) is not a matter of taste: with rows indexed downwards along a cylindric shape the row lengths *increase*, so the correct bead coordinate is $R_i + i$ and not $R_i - i$. I fixed it not by guessing but by rendering WZZ’s own Figure 5.1 ($n = 5$, $m = 2$, weight $(1, 2, 1, 2)$) and reading the box positions off the picture; see §10.

Remark 2.4 (The indexing carries Postnikov’s shift). It is essential that the bead *indexing*, and not merely the bead *set*, is part of the data. The set $A(P)$ alone determines P only up to translation along the diagonal $y = x$; the extra information in $i \mapsto x_i$ is exactly the position of P on that diagonal. Accordingly, replacing (λ_i) by (λ_{i+1}) — the same bead set, relabelled — translates λ diagonally by one and changes $|\lambda/\mu|$ by n : this is Postnikov’s shift parameter d in the notation $\lambda/d/\mu$. A cylindric skew shape is therefore a pair of elements of $\mathcal{C}^{n,m}$ with $\mu \subseteq \lambda$, and letting λ run over all relabellings recovers the whole family $\lambda/d/\mu$. Everything below is stated for a fixed pair, i.e. for a fixed shift.

Remark 2.5. Proposition 2.2 is the only point at which conventions could go wrong, and it is the point at which an error would be invisible — one would prove a correct theorem about the wrong object. It is therefore checked twice computationally in §10: once against an independently coded ordinary skew Kostka computation (the “unrolled” limit $n \gg 0$), and once against WZZ’s own published Example 4.2, which the model reproduces exactly including its negative Schur coefficient.

For the record we also note the bead description of a horizontal strip, since it is what distinguishes the present move from the ribbon move of [4].

Lemma 2.6. $\nu \ll \rho$ if and only if the closed intervals $[\nu_i, \rho_i]$, $i \in \mathbb{Z}$, are pairwise disjoint. Equivalently: the beads move to the right, and no bead’s trajectory contains another bead’s initial or final position.

Proof. $[\nu_i, \rho_i]$ and $[\nu_{i+1}, \rho_{i+1}]$ are disjoint iff $\rho_i < \nu_{i+1}$, since $\nu_i < \nu_{i+1}$ and $\rho_i < \rho_{i+1}$; and the intervals are increasing in i , so pairwise disjointness reduces to consecutive disjointness. $\square$

2.2 Tableaux as chains of bead configurations

By Proposition 2.2, a semistandard cylindric skew tableau of shape λ/μ with ℓ letters is exactly a sequence $x^0, x^1, \dots, x^\ell \in \mathcal{C}^{n,m}$ with $x^0 = \mu$, $x^\ell = \lambda$, and $x^{t-1} \prec\prec x^t$ for $1 \leq t \leq \ell$; its weight is

$$\alpha_t = u(x^t) - u(x^{t-1}), \quad 1 \leq t \leq \ell. \quad (3)$$

Write $\mathcal{T}_\ell(\lambda/\mu)$ for the set of such chains and

$$W_\ell(\lambda/\mu) = \{\alpha(T) : T \in \mathcal{T}_\ell(\lambda/\mu)\} = \text{supp}(s_{\lambda/\mu}^c(x_1, \dots, x_\ell)) \subseteq \mathbb{N}^\ell.$$

Every element of W_ℓ has coordinate sum $d := |\lambda/\mu| = u(\lambda) - u(\mu)$.

3 The decoupling lemma

Throughout this section $p, r \in \mathcal{C}^{n,m}$ are fixed. Define, for $i \in \mathbb{Z}$,

$$L_i = \max(p_i, r_{i-1} + 1), \quad U_i = \min(p_{i+1} - 1, r_i). \quad (4)$$

Both are n -periodic in the sense $L_{i+m} = L_i + n$, $U_{i+m} = U_i + n$.

Lemma 3.1 (Decoupling). *$\{x \in \mathcal{C}^{n,m} : p \prec\prec x \text{ and } x \prec\prec r\} = \{x : \mathbb{Z} \rightarrow \mathbb{Z} \text{ with } x_{i+m} = x_i + n \text{ and } L_i \leq x_i \leq U_i \forall i\}$. In particular this set is a product of integer intervals, and it is nonempty if and only if $L_i \leq U_i$ for all i .*

Proof. $p \prec\prec x$ reads $p_i \leq x_i < p_{i+1}$, i.e. $p_i \leq x_i \leq p_{i+1} - 1$. And $x \prec\prec r$ reads $x_i \leq r_i < x_{i+1}$; the second half, reindexed, is $x_i > r_{i-1}$, i.e. $x_i \geq r_{i-1} + 1$. Together these are exactly $L_i \leq x_i \leq U_i$, and they constrain x_i only through p and r — never through x_j for $j \neq i$.

It remains to check that any n -periodic x with $L_i \leq x_i \leq U_i$ is automatically strictly increasing, hence a cylindric shape. Indeed $x_i \leq U_i \leq p_{i+1} - 1 < p_{i+1} \leq L_{i+1} \leq x_{i+1}$. $\square$

Lemma 3.2 (The palindrome identity). *Suppose $L_i \leq U_i$ for all i . Then*

$$\sum_{i=1}^m (L_i + U_i) = \sum_{i=1}^m (p_i + r_i).$$

Proof. Reindex the U -sum by $j = i + 1$:

$$\sum_{i=1}^m U_i = \sum_{j=2}^{m+1} \min(p_j - 1, r_{j-1}).$$

The $j = m + 1$ term is $\min(p_{m+1} - 1, r_m) = \min(p_1 + n - 1, r_0 + n) = \min(p_1 - 1, r_0) + n$, which is the $j = 1$ term plus n . Hence

$$\sum_{i=1}^m U_i = \sum_{j=1}^m \min(p_j - 1, r_{j-1}) + n = \sum_{j=1}^m (\min(p_j, r_{j-1} + 1) - 1) + n.$$

Since $\max(a, b) + \min(a, b) = a + b$,

$$\sum_{j=1}^m (L_j + U_j) = \sum_{j=1}^m (p_j + r_{j-1} + 1) - m + n = \sum_{j=1}^m p_j + \sum_{j=1}^m r_{j-1} + n.$$

Finally $\sum_{j=1}^m r_{j-1} = \sum_{j=0}^{m-1} r_j = \sum_{j=1}^m r_j - r_m + r_0 = u(r) - n$, and the claim follows. $\square$

Remark 3.3. The term $+n$ produced by the wrap-around is exactly what makes the identity hold; a computation that forgets it yields $\sum(L_i + U_i) = u(p) + u(r) - n$, and with it the false conclusion that the cylindric analogue of Bender–Knuth fails.

Now fix a chain $T = (x^0, \dots, x^\ell) \in \mathcal{T}_\ell(\lambda/\mu)$ and an index $1 \leq t \leq \ell - 1$. Apply (4) with $p = x^{t-1}$ and $r = x^{t+1}$, and let

$$N = \alpha_t + \alpha_{t+1} = u(r) - u(p).$$

By Lemma 3.1 the admissible replacements for x^t form the box $\prod_i [L_i, U_i]$, and by (3) the resulting value of α_t is $u(x^t) - u(p)$, which ranges over *every* integer in

$$[\mu_-, \mu_+], \quad \mu_- = \sum_{i=1}^m L_i - u(p), \quad \mu_+ = \sum_{i=1}^m U_i - u(p),$$

since u is a sum of independently varying integer coordinates. Lemma 3.2 says precisely

$$\mu_- + \mu_+ = N. \tag{5}$$

Corollary 3.4 (Cylindric Bender–Knuth; Postnikov’s symmetry theorem). *The map ι_t which replaces x_i^t by $L_i + U_i - x_i^t$ for all i , and leaves all other rows unchanged, is an involution of $\mathcal{T}_\ell(\lambda/\mu)$ which interchanges α_t and α_{t+1} and fixes all other α_s . Consequently $s_{\lambda/\mu}^c$ is a symmetric function, and $W_\ell(\lambda/\mu)$ is stable under S_ℓ .*

Proof. First, the reflected row is again n -periodic, which is part of what Lemma 3.1 requires of it: since $L_{i+m} = L_i + n$ and $U_{i+m} = U_i + n$, we have $L_{i+m} + U_{i+m} = L_i + U_i + 2n$, so

$$(L_{i+m} + U_{i+m} - x_{i+m}^t) = (L_i + U_i + 2n) - (x_i^t + n) = (L_i + U_i - x_i^t) + n.$$

Granted that, reflection in a box is an involution of the box, so by Lemma 3.1 the result is again a chain (the rows x^s , $s \neq t$, are untouched, and only the steps t and $t+1$ involve x^t). Its weight in position t is

$$\sum_{i=1}^m (L_i + U_i - x_i^t) - u(p) \stackrel{3.2}{=} u(p) + u(r) - u(x^t) - u(p) = u(r) - u(x^t) = \alpha_{t+1},$$

and since $\alpha_t + \alpha_{t+1}$ is unchanged, position $t+1$ receives α_t . The involutions $\iota_1, \dots, \iota_{\ell-1}$ realise the adjacent transpositions of S_ℓ on weights, which generate S_ℓ . $\square$

Corollary 3.5 (The exchange move). *Let $\alpha \in W_\ell(\lambda/\mu)$ and let $1 \leq t \leq \ell - 1$ satisfy $\alpha_t > \alpha_{t+1}$. Then $\alpha - e_t + e_{t+1} \in W_\ell(\lambda/\mu)$. Moreover the move is realised by sliding a single bead of the row x^t one site to the left.*

Proof. Let T be a chain with weight α and form the box of Lemma 3.1 at position t . If $x_i^t = L_i$ for every i then $\alpha_t = \mu_-$, and (5) gives $\mu_- \leq N/2$, i.e. $\alpha_t \leq N - \alpha_t = \alpha_{t+1}$, contrary to hypothesis. Hence $x_j^t > L_j$ for some $j \in \{1, \dots, m\}$. Replacing x_j^t by $x_j^t - 1$ (and, by periodicity, x_{j+km}^t by $x_{j+km}^t - 1$ for all k) stays inside the box, hence gives a chain, by Lemma 3.1; its weight is $\alpha - e_t + e_{t+1}$. $\square$

4 The weight set is a dominance ideal

For $\alpha \in \mathbb{N}^\ell$ let $\text{sort}(\alpha)$ be the partition obtained by sorting its positive entries decreasingly, and let $P(J) = \{\text{sort}(\alpha) : \alpha \in J\}$ for $J \subseteq \mathbb{N}^\ell$. Recall $\sigma \leq \nu$ (dominance) means $\sigma_1 + \dots + \sigma_r \leq \nu_1 + \dots + \nu_r$ for all r , for partitions of the same size.

Lemma 4.1 (Robin Hood step). *Let $\nu \triangleright \sigma$ be partitions of d . Then there are indices $a < b$ with $\nu_a \geq \nu_b + 2$ such that $\tau := \nu - e_a + e_b$ is again a partition and $\nu \triangleright \tau \supseteq \sigma$.*

Proof. Let i be minimal with $\nu_i > \sigma_i$ (it exists and $\nu_c = \sigma_c$ for $c < i$). Let j be minimal with $j > i$ and $\nu_j < \sigma_j$; such j exists, for otherwise $\nu_c \geq \sigma_c$ for all $c \geq i$, which together with $\nu_i > \sigma_i$ contradicts $|\nu| = |\sigma|$. Then $\nu_i > \sigma_i \geq \sigma_j > \nu_j$, so $\nu_i \geq \nu_j + 2$.

Writing $N_r = \nu_1 + \dots + \nu_r$ and $S_r = \sigma_1 + \dots + \sigma_r$, we claim $S_r < N_r$ for $i \leq r < j$. For $r = i$: $S_i = S_{i-1} + \sigma_i = N_{i-1} + \sigma_i < N_{i-1} + \nu_i = N_i$. For $i < r < j$: by minimality of j we have $\sigma_c \leq \nu_c$ for $i < c < j$, so $S_r \leq S_i + \sum_{c=i+1}^r \nu_c \leq (N_i - 1) + \sum_{c=i+1}^r \nu_c = N_r - 1$.

Now set $a = \max\{c : i \leq c < j, \nu_c = \nu_i\}$ and $b = \min\{c : i < c \leq j, \nu_c = \nu_j\}$. Since $\nu_i > \nu_j$ these lie in different constancy blocks of ν , so $a < b$, and $\nu_a = \nu_i \geq \nu_j + 2 = \nu_b + 2$. By the choice of a and b we have $\nu_{a+1} < \nu_a$ and $\nu_{b-1} > \nu_b$, so $\tau = \nu - e_a + e_b$ is weakly decreasing, i.e. a partition (if $b = a + 1$, then $\nu_a - 1 \geq \nu_b + 1$ and this is again clear). Its partial sums are $T_r = N_r - [a \leq r < b]$, and $[a, b) \subseteq [i, j)$, so $S_r \leq T_r$ for all r by the claim. Hence $\sigma \leq \tau \triangleleft \nu$. $\square$

Proposition 4.2. *$P(W_\ell(\lambda/\mu))$ is an order ideal of the dominance order on partitions of d with at most ℓ parts.*

Proof. Let $\nu \in P(W_\ell)$ and $\sigma \leq \nu$; we show $\sigma \in P(W_\ell)$. By Lemma 4.1 and induction on the (finite) number of steps, it suffices to treat $\sigma = \tau = \nu - e_a + e_b$ with $a < b$ and $\nu_a \geq \nu_b + 2$.

By Corollary 3.4, W_ℓ is S_ℓ -stable, so it contains a vector β whose entries are those of ν (padded with zeros) arranged so that $\beta_t = \nu_a$ and $\beta_{t+1} = \nu_b$ for some t , the remaining entries being the remaining parts of ν in any order. Since $\beta_t = \nu_a > \nu_b = \beta_{t+1}$, Corollary 3.5 gives $\beta - e_t + e_{t+1} \in W_\ell$, and $\text{sort}(\beta - e_t + e_{t+1}) = \tau$. $\square$

5 The greedy chain and the dominance maximum

Definition 5.1. The *greedy chain* for λ/μ is defined by $g^0 = \mu$ and

$$g_i^t = \min(g_{i+1}^{t-1} - 1, \lambda_i), \quad i \in \mathbb{Z}, t \geq 1. \quad (6)$$

Lemma 5.2. *Parts (1) and (3) hold unconditionally; part (2) assumes $\mathcal{T}_\ell(\lambda/\mu) \neq \emptyset$.*

1. each g^t is a cylindric shape with $g^{t-1} \nwarrow g^t$ and $g^t \subseteq \lambda$;
2. $g^\ell = \lambda$, so $G := (g^0, \dots, g^\ell) \in \mathcal{T}_\ell(\lambda/\mu)$;
3. for every $T = (x^0, \dots, x^\ell) \in \mathcal{T}_\ell(\lambda/\mu)$ and every t one has $x_i^t \leq g_i^t$ for all i ; in particular $u(x^t) \leq u(g^t)$.

Proof. (1) g^t is n -periodic and strictly increasing, being a coordinatewise minimum of two strictly increasing n -periodic sequences. We induct on t ; the base case is $g^0 = \mu \subseteq \lambda$, which holds by the definition of a cylindric skew shape (Remark 2.4). Assuming inductively $g^{t-1} \subseteq \lambda$ we get

$g_i^{t-1} \leq g_{i+1}^{t-1} - 1$ and $g_i^{t-1} \leq \lambda_i$, so $g_i^{t-1} \leq g_i^t$; and $g_i^t \leq g_{i+1}^{t-1} - 1 < g_{i+1}^{t-1}$, so $g^{t-1} \ll g^t$ by (1). Also $g_i^t \leq \lambda_i$.

(3) Let $T \in \mathcal{T}_\ell$. Then $x^t \subseteq x^\ell = \lambda$ for all t . We induct: $x^0 = g^0$; and if $x_i^{t-1} \leq g_i^{t-1}$ for all i , then $x^t \ll$ -condition gives $x_i^t \leq x_{i+1}^{t-1} - 1 \leq g_{i+1}^{t-1} - 1$, while $x_i^t \leq \lambda_i$; hence $x_i^t \leq g_i^t$.

(2) By (3) applied to any $T \in \mathcal{T}_\ell$ (nonempty by hypothesis), $\lambda_i = x_i^\ell \leq g_i^\ell \leq \lambda_i$. $\square$

Definition 5.3. Let $\gamma_t = u(g^t) - u(g^{t-1})$ be the weight of the greedy chain and $\hat{\lambda} = \hat{\lambda}(\lambda/\mu, \ell) = \text{sort}(\gamma)$.

Proposition 5.4. Let $\ell_0 = \min\{t \geq 0 : g^t = \lambda\}$ (finite whenever $\mathcal{T}_\ell(\lambda/\mu) \neq \emptyset$ for some ℓ). Then $\mathcal{T}_\ell(\lambda/\mu) \neq \emptyset$ if and only if $\ell \geq \ell_0$, and for every such ℓ

$$\hat{\lambda}(\lambda/\mu, \ell) = \hat{\lambda}(\lambda/\mu, \ell_0),$$

i.e. $\hat{\lambda}$ is independent of ℓ on its whole domain of definition.

Proof. The recursion (6) does not involve ℓ , so the sequence $(g^t)_{t \geq 0}$ is one sequence, the same for every ℓ . It is weakly increasing: the proof of Lemma 5.2(1) shows $g_i^{t-1} \leq g_i^t$ for all i . If $g^t = \lambda$ then $g_i^{t+1} = \min(\lambda_{i+1} - 1, \lambda_i) = \lambda_i$, because a cylindric shape is strictly increasing, so $\lambda_i \leq \lambda_{i+1} - 1$; hence $g^s = \lambda$ for all $s \geq \ell_0$.

If $\mathcal{T}_\ell \neq \emptyset$ then $g^\ell = \lambda$ by Lemma 5.2(2), so $\ell \geq \ell_0$. Conversely if $\ell \geq \ell_0$ then $(g^0, \dots, g^\ell)$ is a chain: the steps $t \leq \ell_0$ are horizontal strips by Lemma 5.2(1), and the steps $t > \ell_0$ are $\lambda \ll \lambda$, which holds by (1) for the same reason $\lambda_i \leq \lambda_{i+1} - 1$. So $\mathcal{T}_\ell \neq \emptyset$.

Finally, for $\ell \geq \ell_0$ the weight of the greedy chain is $\gamma = (\gamma_1, \dots, \gamma_{\ell_0}, 0, \dots, 0)$, since $\gamma_t = u(g^t) - u(g^{t-1}) = 0$ for $t > \ell_0$. Sorting discards the trailing zeros, so $\text{sort}(\gamma)$ is the same partition for every $\ell \geq \ell_0$. $\square$

Proposition 5.5. $\hat{\lambda} \in P(W_\ell)$ and $\nu \leq \hat{\lambda}$ for every $\nu \in P(W_\ell)$.

Proof. $\gamma \in W_\ell$ by Lemma 5.2(2), and $\hat{\lambda} = \text{sort}(\gamma) \in W_\ell$ by Corollary 3.4; so $\hat{\lambda} \in P(W_\ell)$.

Let $\nu \in P(W_\ell)$ and $r \geq 1$. Since W_ℓ is S_ℓ -stable, ν itself (padded with zeros) lies in W_ℓ , say $\nu = \alpha(T)$ for some $T = (x^0, \dots, x^\ell)$. Telescoping (3),

$$\nu_1 + \dots + \nu_r = u(x^r) - u(x^0) \stackrel{5.2(3)}{\leq} u(g^r) - u(g^0) = \gamma_1 + \dots + \gamma_r \leq \hat{\lambda}_1 + \dots + \hat{\lambda}_r,$$

the last inequality because $\hat{\lambda}_1 + \dots + \hat{\lambda}_r$ is the maximum of $\gamma_{t_1} + \dots + \gamma_{t_r}$ over all r -subsets. Hence $\nu \leq \hat{\lambda}$. $\square$

6 From a dominance ideal with a maximum to M-convexity

This section is self-contained polymatroid theory; we include it so that the proof cites nothing.

Lemma 6.1. Let $\rho : 2^{[\ell]} \rightarrow \mathbb{Z}$ be submodular and let $J = \{\alpha \in \mathbb{N}^\ell : \alpha([\ell]) = \rho([\ell]), \alpha(S) \leq \rho(S) \forall S\}$, where $\alpha(S) = \sum_{i \in S} \alpha_i$. For $\alpha \in J$ the family $\mathcal{F}_\alpha = \{S : \alpha(S) = \rho(S)\}$ is closed under union and intersection.

Proof. For $S, T \in \mathcal{F}_\alpha$, $\rho(S) + \rho(T) = \alpha(S) + \alpha(T) = \alpha(S \cup T) + \alpha(S \cap T) \leq \rho(S \cup T) + \rho(S \cap T) \leq \rho(S) + \rho(T)$, forcing equality throughout. $\square$

Lemma 6.2 (sorted form = subset form). *Let $\hat{\lambda}$ be a partition of d with at most ℓ parts and $\Lambda_r = \hat{\lambda}_1 + \dots + \hat{\lambda}_r$. Then*

$$\{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \leq \hat{\lambda}\} = \{\alpha \in \mathbb{N}^\ell : \alpha([\ell]) = d, \alpha(S) \leq \Lambda_{|S|} \forall S \subseteq [\ell]\}.$$

Proof. For $\alpha \in \mathbb{N}^\ell$ and $0 \leq r \leq \ell$,

$$\max_{|S|=r} \alpha(S) = \text{sort}(\alpha)_1 + \dots + \text{sort}(\alpha)_r, \quad (7)$$

with $\text{sort}(\alpha)_k = 0$ beyond the number of positive entries: taking S to index r largest entries realises the right-hand side, and any $\alpha(S)$ with $|S| = r$ is a sum of r entries, hence at most the sum of the r largest.

($\subseteq$) Let $\text{sort}(\alpha) \leq \hat{\lambda}$. Dominance is defined only between partitions of the *same size*, so this includes $\alpha([\ell]) = |\text{sort}(\alpha)| = |\hat{\lambda}| = d$; and for $|S| = r$, (7) gives $\alpha(S) \leq \text{sort}(\alpha)_1 + \dots + \text{sort}(\alpha)_r \leq \Lambda_r$.

($\supseteq$) Let $\alpha([\ell]) = d$ and $\alpha(S) \leq \Lambda_{|S|}$ for all S . Then $|\text{sort}(\alpha)| = d = |\hat{\lambda}|$, so dominance between them is meaningful. For $1 \leq r \leq \ell$ pick, by (7), an S with $|S| = r$ and $\alpha(S) = \text{sort}(\alpha)_1 + \dots + \text{sort}(\alpha)_r$; then this partial sum is $\leq \Lambda_r$. For $r > \ell$ both partial sums equal d . Hence $\text{sort}(\alpha) \leq \hat{\lambda}$. $\square$

Remark. Both inclusions use the equal-size convention, in opposite directions; it is the only content the one-line justification used to elide. Without it ($\supseteq$) fails: for $\ell = 2$, $\hat{\lambda} = (2, 1)$, $\alpha = (1, 1)$ satisfies every inequality $\alpha(S) \leq \Lambda_{|S|}$ but is not dominance-comparable to $\hat{\lambda}$ at all. Lemma 6.2 was also the last step of Proposition 6.3 to be formalised in Lean — it is machine-checked, sorry-free, as of 25 September 2026 (Remark 6.4) — and it is the hypothesis of Theorem 8.1; it is load-bearing in two independent places, which is why it is now a lemma with a proof instead of a clause.

Proposition 6.3. *Let $\hat{\lambda}$ be a partition of d with at most ℓ parts and set $\Lambda_r = \hat{\lambda}_1 + \dots + \hat{\lambda}_r$ for $0 \leq r \leq \ell$. Then*

$$J = \{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \leq \hat{\lambda}\} = \{\alpha \in \mathbb{N}^\ell : \alpha([\ell]) = d, \alpha(S) \leq \Lambda_{|S|} \forall S \subseteq [\ell]\}$$

is M -convex.

Proof. The displayed equality is Lemma 6.2. Put $\rho(S) = \Lambda_{|S|}$. As $\hat{\lambda}$ is weakly decreasing, $r \mapsto \Lambda_r$ is concave; and for any S, T we have $|S \cup T| + |S \cap T| = |S| + |T|$ with $|S \cap T| \leq \min(|S|, |T|) \leq \max(|S|, |T|) \leq |S \cup T|$, so concavity gives $\Lambda_{|S \cup T|} + \Lambda_{|S \cap T|} \leq \Lambda_{|S|} + \Lambda_{|T|}$: ρ is submodular.

Let $\alpha, \beta \in J$ and i with $\alpha_i > \beta_i$. Note $\alpha - e_i + e_j \in \mathbb{N}^\ell$ for any j , since $\alpha_i \geq 1$. For $S \subseteq [\ell]$, $(\alpha - e_i + e_j)(S) = \alpha(S) - [i \in S] + [j \in S]$ exceeds $\rho(S)$ only if $j \in S$, $i \notin S$ and $S \in \mathcal{F}_\alpha$. Hence

$$\alpha - e_i + e_j \notin J \iff \exists S \in \mathcal{F}_\alpha \text{ with } j \in S, i \notin S.$$

Let $D = \{j : \alpha_j < \beta_j\}$; $D \neq \emptyset$ because $\alpha([\ell]) = \beta([\ell])$ and $\alpha_i > \beta_i$, and $i \notin D$. Suppose, for contradiction, that $\alpha - e_i + e_j \notin J$ for every $j \in D$, and pick $S_j \in \mathcal{F}_\alpha$ with $j \in S_j$, $i \notin S_j$. By Lemma 6.1, $S^* = \bigcup_{j \in D} S_j \in \mathcal{F}_\alpha$, and $D \subseteq S^*$, $i \notin S^*$. Now count on the *complement* $C = [\ell] \setminus S^*$. Since $D \subseteq S^*$, no index of D lies in C , so $\beta_c \leq \alpha_c$ for every $c \in C$; and $i \in C$ with $\beta_i < \alpha_i$ strictly. Hence $\beta(C) < \alpha(C)$, and since $\alpha([\ell]) = \beta([\ell]) = d$,

$$\beta(S^*) = d - \beta(C) > d - \alpha(C) = \alpha(S^*) = \rho(S^*),$$

contradicting $\beta \in J$. $\square$

Remark 6.4 (Erratum, 25 September 2026; and what is machine-checked). The last step of the proof above previously ran the count on S^* itself, via $\beta(S^*) - \alpha(S^*) = \sum_{j \in S^*} (\beta_j - \alpha_j) \geq \sum_{j \in D} (\beta_j - \alpha_j) > 0$. That inequality is the wrong way round: the stated reason — every term with $j \in S^* \setminus D$ is ≤ 0 — gives $\leq$, not $\geq$. The conclusion is unaffected; the count simply has to be made on the complement, as it now is, using only ingredients the proof had already established ($D \subseteq S^*$, $i \notin S^*$, $\alpha([\ell]) = \beta([\ell])$). The slip was found by the Lean type checker refusing the original chain.

Lemma 6.1 and Proposition 6.3 are now formalised, sorry-free, in `TworowD4Kernel/MConvexExchange.lean` of `github.com/clio-vega/tworow-d4-kernel` (`MConvexExchange.tight_union`, `MConvexExchange.insupp_exchange`) axioms exactly [`propext`, `Classical.choice`, `Quot.sound`]). The sorted/subset identification is no longer a gap. J is formalised in the subset form $\{\alpha : \alpha([\ell]) = d, \alpha(S) \leq \Lambda_{|S|}\}$, and its identification with the sorted form $\text{sort}(\alpha) \leq \hat{\lambda}$ — the first sentence of the proof above, now Lemma 6.2 — is itself formalised, sorry-free, as `SortedBridge.insupp_iff_sorted`, together with the exchange axiom for the sorted form itself (`SortedBridge.sorted_exchange`), in `TworowD4Kernel/SortedSubsetBridge.lean` (commits `cc0573b`, `85054bb`). The definition of `sort` is not chosen to make the theorem true: it is composition with an explicit sorting permutation of the index list, and the two properties that pin a sorted rearrangement down uniquely (`sortDesc_antitone`, `sortDesc_multiset`) are theorems, so a wrong definition is falsifiable inside Lean. An independent enumeration agrees (0 disagreements over all 1804 pairs $(\hat{\lambda}, \ell)$ with $|\hat{\lambda}| \leq 16$, $\ell \leq 6$).

One caveat does remain, stated because the type checker cannot see past it: what is proved is the one-sided exchange axiom, which is the definition of M-convexity used in §1.1 and by WZZ; Murota’s symmetric axiom, which demands $\beta + e_i - e_j \in J$ for the *same* j , is equivalent by a theorem of Murota–Shioura that is neither used nor formalised here, and is checked by brute force instead (0 failures over 876317 triples (α, β, i) in the same range). The code is `projects/proofs/code-q254-lean/mconvex_exchange_check.py`.

7 The main theorem

Theorem 7.1. *Let λ/μ be a cylindric skew shape in $C^{n,m}$, let $\ell \geq 1$, and suppose $\mathcal{T}_\ell(\lambda/\mu) \neq \emptyset$. Let $\hat{\lambda} = \hat{\lambda}(\lambda/\mu, \ell)$ be the sorted weight of the greedy chain of Definition 5.1. Then*

$$\text{supp}(s_{\lambda/\mu}^c(x_1, \dots, x_\ell)) = W_\ell(\lambda/\mu) = \{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \leq \hat{\lambda}\} = \mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^\ell,$$

and this set is M-convex. Consequently $s_{\lambda/\mu}^c(x_1, \dots, x_\ell)$ is M-convex, has saturated Newton polytope, and $\text{Newton}(s_{\lambda/\mu}^c) = \mathcal{P}_{\hat{\lambda}}$.

Proof. By Proposition 5.5, every $\nu \in P(W_\ell)$ satisfies $\nu \leq \hat{\lambda}$, and $\hat{\lambda} \in P(W_\ell)$. By Proposition 4.2, $P(W_\ell)$ is a dominance ideal, so it contains every $\sigma \leq \hat{\lambda}$. Hence $P(W_\ell) = \{\sigma : \sigma \leq \hat{\lambda}\}$, and since W_ℓ is S_ℓ -stable (Corollary 3.4), $W_\ell = \{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \leq \hat{\lambda}\}$, which is the second equality. M-convexity now follows from Proposition 6.3, and does not use anything below. SNP is Theorem 8.1, the Newton polytope identification is Theorem 8.4, and the third equality $W_\ell = \mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^\ell$ is the last sentence of Theorem 8.4; all three are proved in §8 below. $\square$

Remark 7.2. The three conclusions are proved in increasing order of cost, and the order matters. M-convexity uses only the second equality. SNP (Theorem 8.1) adds Lemma 6.2 and the convexity of a half-space intersection, and never mentions $\mathcal{P}_{\hat{\lambda}}$. Only the Newton identification

(Theorem 8.4) needs the majorisation argument of Proposition 8.3. Earlier drafts of this proof derived SNP from $\text{conv}(W_\ell) = \mathcal{P}_{\hat{\lambda}}$, which had the dependency the wrong way round and made SNP look as though it rested on Rado's theorem; it does not.

Remark 7.3. If $\mathcal{T}_\ell(\lambda/\mu) = \emptyset$ then $s_{\lambda/\mu}^c(x_1, \dots, x_\ell) = 0$ and there is nothing to prove. The greedy chain also computes the minimal ℓ for which $\mathcal{T}_\ell \neq \emptyset$: it is the least t with $g^t = \lambda$.

8 Saturated Newton polytope and the Newton polytope

This section proves the last two assertions of Theorem 7.1. It uses from above only Lemma 6.2, the second equality of Theorem 7.1, the Robin Hood step (Lemma 4.1) and S_ℓ -stability (Corollary 3.4). A longer discussion, including the computational verification, is in the companion note 2026-09-25-c2-newton-polytope-and-snp.

Theorem 8.1 (SNP). $\text{conv}(W_\ell) \cap \mathbb{Z}^\ell = W_\ell$; that is, $s_{\lambda/\mu}^c(x_1, \dots, x_\ell)$ has saturated Newton polytope.

Proof. Let $Q = \{x \in \mathbb{R}^\ell : x_i \geq 0 \ \forall i, x([\ell]) = d, x(S) \leq \Lambda_{|S|} \ \forall S \subseteq [\ell]\}$, an intersection of finitely many closed half-spaces with a hyperplane, hence convex; and $Q \cap \mathbb{Z}^\ell$ is exactly the subset-form description of Lemma 6.2, since an integer vector with all entries ≥ 0 lies in $\mathbb{N}^\ell$. By Theorem 7.1 and Lemma 6.2, $W_\ell = Q \cap \mathbb{Z}^\ell$; in particular $W_\ell \subseteq Q$, so $\text{conv}(W_\ell) \subseteq Q$ because Q is convex. Therefore

$$\text{conv}(W_\ell) \cap \mathbb{Z}^\ell \subseteq Q \cap \mathbb{Z}^\ell = W_\ell,$$

and the reverse inclusion is trivial. $\square$

No permutahedron and no majorisation theorem entered that proof. The next one is different.

Lemma 8.2 (the segment step). Let $\nu \in \mathbb{R}^\ell$ and $a \neq b$ with $g := \nu_a - \nu_b \geq 1$, and let $\tau = (a \ b) \cdot \nu$. Then $\nu - e_a + e_b = (1 - \frac{1}{g})\nu + \frac{1}{g}\tau$, a convex combination of ν and τ .

Proof. $\tau - \nu = g(e_b - e_a)$, so $\nu + \frac{1}{g}(\tau - \nu) = \nu - e_a + e_b$; and $\frac{1}{g} \in (0, 1]$. $\square$

Proposition 8.3. Let $K \subseteq \mathbb{R}^\ell$ be convex and stable under the coordinate action of S_ℓ , and let $\hat{\lambda} \in K$ be a partition of d with at most ℓ parts. Then $\{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \preceq \hat{\lambda}\} \subseteq K$. Consequently $\mathcal{P}_{\hat{\lambda}}$, which is the smallest such K , contains that set.

Proof. Since α is a permutation of $\sigma := \text{sort}(\alpha)$ (padded to length ℓ) and K is S_ℓ -stable, it suffices to show $\sigma \in K$ for every partition $\sigma \preceq \hat{\lambda}$ of d with at most ℓ parts.

Set $\nu^{(0)} = \hat{\lambda}$ and, while $\nu^{(t)} \neq \sigma$, use Lemma 4.1 — legitimate because the invariant $\nu^{(t)} \trianglerighteq \sigma$ together with $\nu^{(t)} \neq \sigma$ gives $\nu^{(t)} \triangleright \sigma$ — to obtain $a_t < b_t$ with $\nu_{a_t}^{(t)} \geq \nu_{b_t}^{(t)} + 2$ and $\nu^{(t+1)} := \nu^{(t)} - e_{a_t} + e_{b_t}$ a partition with $\nu^{(t)} \triangleright \nu^{(t+1)} \trianglerighteq \sigma$, so the invariant persists. Here $b_t \leq \ell$: in the proof of Lemma 4.1 one has $b \leq j$ with $\nu_j^{(t)} < \sigma_j$, and $\sigma_j > 0$ forces $j \leq \ell$ as σ has at most ℓ parts. Also $\nu^{(t+1)} \in \mathbb{N}^\ell$, since $\nu_{a_t}^{(t+1)} = \nu_{a_t}^{(t)} - 1 \geq \nu_{b_t}^{(t)} + 1 \geq 1$.

Termination. With $N_r^{(t)}, S_r$ the partial sums of $\nu^{(t)}, \sigma$, put $\Phi(t) = \sum_{r=1}^\ell (N_r^{(t)} - S_r) \in \mathbb{N}$ (each summand is ≥ 0 by the invariant). The partial sums of $\nu^{(t+1)}$ are $N_r^{(t)} - [a_t \leq r < b_t]$, so $\Phi(t+1) = \Phi(t) - (b_t - a_t) \leq \Phi(t) - 1$, using $a_t < b_t \leq \ell$. A strictly decreasing sequence in $\mathbb{N}$ is finite, so the recursion halts, and it halts only at $\nu^{(k)} = \sigma$ with $k \leq \Phi(0)$.

The chain stays in K . By induction: $\nu^{(0)} = \hat{\lambda} \in K$; and if $\nu := \nu^{(t)} \in K$ then $(a_t b_t) \cdot \nu \in K$ by S_ℓ -stability, so by Lemma 8.2 (applicable as $\nu_{a_t} - \nu_{b_t} \geq 2 \geq 1$) the point $\nu^{(t+1)}$ is a convex combination of two points of K , hence in K . Thus $\sigma = \nu^{(k)} \in K$.

Finally $\mathcal{P}_{\hat{\lambda}} = \text{conv}(S_\ell \cdot \hat{\lambda})$ is convex, is S_ℓ -stable (the action is linear, so $w \cdot \text{conv}(S_\ell \hat{\lambda}) = \text{conv}(S_\ell \hat{\lambda})$), and contains $\hat{\lambda}$; and any convex S_ℓ -stable $K \ni \hat{\lambda}$ contains $S_\ell \cdot \hat{\lambda}$ and hence $\mathcal{P}_{\hat{\lambda}}$. $\square$

Theorem 8.4 (Newton polytope). $\text{conv}(W_\ell) = \mathcal{P}_{\hat{\lambda}}$, i.e. $\text{Newton}(s_{\lambda/\mu}^c(x_1, \dots, x_\ell)) = \mathcal{P}_{\hat{\lambda}}$. Moreover $W_\ell = \mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^\ell$.

Proof. $\mathcal{P}_{\hat{\lambda}} \subseteq \text{conv}(W_\ell)$: $\hat{\lambda} \in W_\ell$ by Theorem 7.1, and W_ℓ is S_ℓ -stable by Corollary 3.4, so $S_\ell \cdot \hat{\lambda} \subseteq W_\ell$ and hence $\mathcal{P}_{\hat{\lambda}} = \text{conv}(S_\ell \cdot \hat{\lambda}) \subseteq \text{conv}(W_\ell)$.

$\text{conv}(W_\ell) \subseteq \mathcal{P}_{\hat{\lambda}}$: by Theorem 7.1, $W_\ell = \{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \leq \hat{\lambda}\}$, which is contained in $\mathcal{P}_{\hat{\lambda}}$ by Proposition 8.3; now take convex hulls.

Finally $W_\ell \subseteq \mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^\ell$, and conversely $\mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^\ell = \text{conv}(W_\ell) \cap \mathbb{Z}^\ell = W_\ell$ by Theorem 8.1. $\square$

Remark 8.5 (This re-proves Rado; it does not avoid him). Proposition 8.3 applied to $K = \mathcal{P}_{\hat{\lambda}}$ is the non-trivial direction of Rado's theorem, equivalently of the Hardy–Littlewood–Pólya majorisation theorem: $\text{sort}(\alpha) \leq \hat{\lambda}$ implies $\alpha \in \text{conv}(S_\ell \cdot \hat{\lambda})$. The proof above is moreover the classical one — Lemma 4.1 is the classical transfer step. So the honest statement is that we *re-prove* Rado's direction from Lemma 4.1, not that we avoid it. What this note claims, and what WZZ Problem 5.2 asks for, is unaffected: the proof *cites* nothing, and Lemma 4.1 was proved here from scratch for a different purpose (Proposition 4.2) and turns out to drive the majorisation argument too. We claim no novelty for Proposition 8.3.

Note also what is *not* proved: nothing above asserts $Q = \mathcal{P}_{\hat{\lambda}}$ for the polyhedron Q of Theorem 8.1. Only $\text{conv}(W_\ell) \subseteq Q$ is used. The equality is the real-coefficient form of Rado's theorem and is needed nowhere here.

9 What the proof consumes

The success criterion attached to WZZ Problem 5.2 is that the argument must not route through [2, Cor. 8.5]. We therefore list every external input.

- **Lam, Cor. 8.5** (affine Stanley $\Rightarrow$ dual k -Schur positivity): *not used*.
- **Lee 2019, Cor. 5** (cylindric skew Schur = affine Stanley for a 321-avoiding affine permutation): *not used*.
- **WZZ Theorem 3.6** (k -Kostka positivity) and **Theorem 4.4**: *not used*.
- **Rado's theorem** ($\mathcal{P}_\mu \subseteq \mathcal{P}_\lambda \iff \mu \leq \lambda$): *not used*. Proposition 6.3 proves M-convexity directly from the exchange axiom.
- **Postnikov's symmetry theorem** [5] for cylindric skew Schur functions: *not used* — it is reproved as Corollary 3.4.
- **Murota's theory of M-convex sets**: only the *definition* is used.
- **Classical facts about partitions**: that λ/μ is a horizontal strip iff $\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \dots$, and the β -set encoding of a partition (Macdonald I.(1.7)). These are used once, inside Proposition 2.2, to translate the lattice-path definition into the bead model.

The proof is therefore combinatorial and based on the tableau interpretation, as Problem 5.2 requests.

Remark 9.1 (What the answer looks like as a tableau statement). Unwinding Corollary 3.5 through Proposition 2.2: given a semistandard cylindric skew tableau in which the letter t occurs more often than the letter $t + 1$ (per period), one may change a single box from t to $t + 1$. The box to change is located by the bead picture: the row λ^t of the chain is not at the left end of its box $\prod_i [L_i, U_i]$, and retracting one bead of λ^t by one site is exactly the required box change. The cylindric Bender–Knuth involution (Corollary 3.4) is the full reflection of that box.

10 Computational verification

All computations were run in Python over the bead model of §2; the code is at [projects/scratch/q191/code/cyl](https://github.com/q191/code/cyl)

Anchor (is the model the right object?). For $n = 20$, $m = 2$, $\mu = (0, 2)$, $\lambda = (3, 7)$ — a cylinder wide enough that the shape does not wrap — the bead model’s weight generating function agrees in all 13 coefficients with an independently coded ordinary skew Kostka computation for the corresponding skew partition shape $(8, 5)/(3, 2)$.

Positive control (WZZ Example 4.2). WZZ record, for $w = s_2 s_1 s_0 s_2 \in \tilde{S}_3$, $\tilde{F}_w = m_{1111} + m_{211} + m_{22} = s_{22} - s_{1111}$. A search over $\mathcal{C}^{3,m}$ ($m = 1, 2$) with $d = 4$ finds exactly one realisation, $m = 1$, $\mu = (0)$, $\lambda = (4)$, and its monomial expansion is $m_{1111} + m_{211} + m_{22}$; converting to the Schur basis gives $s_{22} - s_{1111}$, reproducing the published value including the negative coefficient. This control is untuned (published by other authors for exactly this purpose), signed, and has M-convex support — so it also certifies that the model is *not* computing ordinary skew Schur functions in disguise.

Second control, and the one that fixed the orientation (WZZ Figure 5.1). WZZ illustrate a semistandard cylindric skew tableau for $n = 5$, $m = 2$, weight $(1, 2, 1, 2)$. I compiled their `ytableau` source to a picture, read the boxes off it, and converted with (2): the shape is $\mu = (0, 2)$, $\lambda = (3, 5)$ in $\mathcal{C}^{5,2}$ — two rows of $3 = n - m$ boxes, 6 boxes per period — and their tableau is the chain $(0, 2) \subseteq (0, 3) \subseteq (1, 4) \subseteq (2, 4) \subseteq (3, 5)$, every step of which the model certifies as a cylindric horizontal strip, of sizes 1, 2, 1, 2 exactly as printed. This is what settled the sign in (2); the naive choice $R_i - i$ does not even give a strictly increasing sequence.

Lemma 3.1 (decoupling). For all $2 \leq n \leq 5$, $1 \leq m < n$, all p with $p_1 = 0$, and all $r \supseteq p$ within a window of 5: the set of admissible middle shapes coincides with $\prod_i [L_i, U_i]$ in all 380 feasible instances (290 middle shapes in total). No failures.

Lemma 3.2 (palindrome identity). 470 feasible (p, r) pairs over $2 \leq n \leq 6$: $\mu_- + \mu_+ = N$ in all 470. (This computation corrected an algebra slip in my first derivation, which dropped the wrap-around term and gave $N - n$.)

Lemma 5.2 and Proposition 5.5 (greedy maximum). 462 instances over $2 \leq n \leq 5$, $d \leq 7$, $\ell \leq 5$: in every one the greedy chain reaches λ in ℓ steps and $\text{sort}(\gamma)$ is a member of $P(W_\ell)$ dominating every element of $P(W_\ell)$. No failures.

Theorem 7.1. 1932 instances over $2 \leq n \leq 6$, $d \leq 7$: $P(W_\ell)$ is a dominance ideal with a unique maximal element in every one. Separately, 409 instances were checked against the raw definition of M -convexity by exhaustive search over all pairs $(\alpha, \beta) \in W_\ell^2$ and all i ; all pass, and in all 409 the support is S_ℓ -stable, confirming Corollary 3.4. Finally, 339 instances were checked for symmetry of the *coefficients* (not merely the support); all pass.

11 Gaps

I record the following honestly.

1. **(Closed.)** An earlier draft proved Proposition 2.2 by local reduction to the classical partition case, and flagged the window bookkeeping as the one step not written at full length. That is no longer the situation: with the coordinate $x_i = R_i + i$ of (2) the three statements are proved directly and in full. Two errors were made and corrected on the way to it, both recorded above: the first draft claimed a bijection with periodic bead *sets* (false — a bead set determines the path only up to diagonal translation, and the lost data is Postnikov’s shift d ; Remark 2.4), and the first draft of the sign used $R_i - i$ (false — row lengths increase downwards; Remark 2.3).
2. **(Closed, 25 September 2026, cycle 2.)** The clause previously printed here read: “ $\mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^\ell = \{\alpha : \text{sort}(\alpha) \preceq \hat{\lambda}\}$ is asserted without proof in Theorem 7.1. It is Rado’s theorem, and it is used only for the cosmetic identification of the Newton polytope; M -convexity, SNP and the description of the support do not depend on it.” The assertion is now proved — §8, Theorems 8.1 and 8.4 — and three things about the way it had been filed were wrong.

First, “cosmetic” was wrong. $\text{Newton}(s_{\lambda/\mu}^c) = \mathcal{P}_{\hat{\lambda}}$ is one of the three conclusions of Theorem 7.1 and is advertised in the abstract. A conclusion a reader is invited to quote is not cosmetic, and grading it so is precisely what ensured that nobody, including the author, ever read the one-clause derivation that rested on it.

Second, the claim that SNP does not depend on it was true but not enacted. SNP is indeed independent of Rado — Theorem 8.1 proves it from Lemma 6.2 and the convexity of a half-space intersection alone — but the proof of Theorem 7.1 as previously printed derived SNP from $\text{conv}(W_\ell) = \mathcal{P}_{\hat{\lambda}}$, i.e. from exactly the unproved statement. The independence was asserted in this list and contradicted by the proof three pages earlier.

Third, “it is Rado’s theorem” was right, and the fix does not avoid Rado — it re-proves him. Proposition 8.3 with $K = \mathcal{P}_{\hat{\lambda}}$ is the non-trivial direction of Rado’s theorem, by the classical argument, whose transfer step is Lemma 4.1. What survives, and is what was actually wanted, is that this note *cites* nothing: Lemma 4.1 is proved here from scratch for an unrelated purpose (Proposition 4.2) and turns out to drive the majorisation argument too. See Remark 8.5, which states this in place of any claim of independence.

Still not proved, and not used: $Q = \mathcal{P}_{\hat{\lambda}}$ for the polyhedron Q of Theorem 8.1, i.e. the real-coefficient form of Rado’s theorem. Only $\text{conv}(W_\ell) \subseteq Q$ is needed and only that is proved. Lemma 6.2 now has a written proof *and* a machine-checked one (Remark 6.4). What is still outside the Lean record is Murota’s *symmetric* exchange axiom, which is checked by brute force only.

3. **Dependence of $\hat{\lambda}$ on ℓ — corrected 23 September 2026, cycle 2.** The clause previously printed here read: “ $\hat{\lambda}(\lambda/\mu, \ell)$ grows with ℓ until ℓ reaches the minimal chain length, after which it is constant. I have not written out the (easy) monotonicity statement.” That clause is **false**, and the monotonicity statement it promised does not exist because there is nothing to be monotone: $\hat{\lambda}$ does not depend on ℓ at all. There is no “growing” regime, because on the range where $\hat{\lambda}$ is undefined the tableau set is empty. The correct statement is Proposition 5.4, now proved in §5. The defect was visible from the definition: the recursion (6) defining the greedy chain contains no ℓ .

References

- [1] Wang, Zhang, Zhang, *Newton polytopes and M-convexity of dual k -Schur polynomials*, arXiv:2401.14632. (Read at LaTeX source; M-convexity definition at line 122, Theorem 3.6, Theorem 4.4, Corollary 4.9, Problems 5.1–5.2 at lines 938 and 945.)
- [2] T. Lam, *Schubert polynomials for the affine Grassmannian*, J. Amer. Math. Soc. 21 (2008). (Cited only to say it is *not* used.)
- [3] S. J. Lee, *Positivity of cylindric skew Schur functions*, J. Combin. Theory Ser. A **168** (2019), 26–49; doi:10.1016/j.jcta.2019.05.009; arXiv:1706.04460. (Cited only to say it is *not* used. Title and coordinates verified 2026-09-23 against the arXiv metadata API and, independently, Crossref; earlier drafts of this bibliography gave the title as “Cylindric skew Schur functions and affine Stanley symmetric functions”, which is not the title of any paper.)
- [4] L. Naprienko, *Free fermionic Schur functions*, arXiv:2301.12110. (Read at LaTeX source; Lemma 4.4 `lem:ribbonmaya` at line 1528, bead/hole conventions at eq. (4.1).)
- [5] A. Postnikov, *Affine approach to quantum Schubert calculus*, Duke Math. J. 128 (2005), no. 3, 473–509. (The source of cylindric skew Schur functions and of the symmetry theorem reproved here as Corollary 3.4. Read at LaTeX source 2026-09-23, arXiv:math/0205165v2, cited by internal label and line number: `cor:K-Sym l.1456`, `prop:s-kappa-symmetric l.1502` (proved from commutativity of the quantum product), the assertion of an unwritten combinatorial proof via pairwise-commuting strip-adding operators at ll.1460–1464, and `lem:e-h-commute-in-AN l.2882–2893` in the affine nil-Temperley–Lieb algebra $\mathcal{A}_n$.)
- [6] I. M. Gessel and C. Krattenthaler, *Cylindric partitions*, Trans. Amer. Math. Soc. 349 (1997), no. 2, 429–479. (Not on arXiv; obtained 2026-09-23 from Krattenthaler’s homepage, `~kratt/artikel/cylindri.html`. Proposition 1 read at source; the remainder skimmed. The PostScript-to-text conversion drops math glyphs, so displayed formulas must be read in the PDF, not the extracted text.)
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- [8] E. Neyman, *Cylindric Young tableaux and their properties*, arXiv:1410.5039. (Read at LaTeX source; Thm `schurpols` at line 1677 and Cor. `oneschurcor` at line 1824 verified computationally in `reviews/2026-09-22-c2-q223-neyman.md`. Contains no statement about supports, polytopes or M-convexity; see that review, §2.)

- [9] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., I.(1.7).